Modeling Scaling of SWRO Membranes in Simscape

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1 The Motivation

Scaling in SWRO is the growth of mineral deposits on the surface of the membrane, reducing the effective membrane area and thus reducing the permeate production. What makes this a particularly compelling addition to the existing transient desalination modeling capabilities is that scaling is a transient phenomenon, and necessitates non constant operating conditions to be properly modeled.

2 Will it scale?

Lots of literature discusses predicting if scaling will happen or not. There are two main variables (very similar) that are used to predict scaling. The Saturation Index (SI) and the Supersaturation Ratio (S_r). Both are functions of the ion activity product (IAP) and the solubility product (K_{sp}). The equations are shown below:

$$SI = log\left(\frac{IAP}{K_{sp}}\right)$$

$$S_r = \sqrt{\frac{IAP}{K_{sp}}}$$

If $SI > 0$ or $S_r > 1$, scaling is likely to occur. Effectively these say the same thing if the ion activity product exceeds the solubility product, scaling is likely to occur.

The ion activity product is a function of the concentration of ions in solution, and can be calculated as follows:

$$IAP = \prod_{i=\text{ions}} \gamma_i [i]$$

where γ_i is the activity coefficient of ion i, and $[i]$ is the concentration of ion i in mol/L. We already have the concentration of ions in kg/m^3 from our solution domain, so we can convert to mol/L with an added fluid property for molar mass (M_x). The activity coefficient and the solubility product can probably be dependent variables in the future, but for now I will treat them as constant fluid parameters.

3 From looking at plots and Nate thinking...

Just knowing if scaling will occur is not sufficient for what we would like to model. This does not help us track performance over time as scaling builds up, or help us understand operational strategies to mitigate scaling. To do this we need to model the rate of scaling. While there are many models for determining if scaling will occur, there are less models for scaling rate. From some plots online, it appears that the growth rate is an exponential function. The scale size vs time plots generally start low, then gow into exponential till it hits a max slope. The growth rate seems to be dependent on 2 things, concentration (or rather SI) and size of existing scale.

From a source, the growth rate appears to take this form:

$$\frac{dM_{\text{scale}}}{dt} = A(1 - e^{-t/\tau})$$

where A is a maximum growth rate, that we see once the scale has some sort of critical mass and τ is a time constant that determines how quickly we reach that max growth rate. This equation makes a lot of sense to me, as growth rate should increase as the scale gets larger (increased area for more particles to attach to), but at some point when the membrane is covered, the growth rate should plateau. But then it should probably also eventually reach a maximum scale mass. So maybe a better model would be:

$$\frac{dM_{\text{scale}}}{dt} = A(1 - e^{-t/\tau})(1 - \frac{M_{\text{scale}}}{M_{\text{max}}})$$

where M_{max} is the maximum mass of scale that can build up on the membrane. This equation captures the initial increase in growth rate as scale builds up, but also captures the eventual plateauing of scale mass as it approaches a maximum value.

Unfortunately, I have not been able to find any sources that use such an equation, most are far too detailed for our purposes. Moving forward I'm going to look into general crystal growth models instead of specifically SWRO scaling models.

4 From Textbook

Rates of crystallization are typically empirical. There are two main stages. The diffusional stage is where solute is transported from the fluid and onto the crystal surface. The deposition stage is where the solutes on the crystal surface integrate into the lattice structure. The textbook describes both stages of growth with:

$$\frac{dm}{dt} = k_d A(c - c_i) = k_r A(c_i - c^*)^i$$

where k_d and k_r are the diffusion and deposition rate or reaction mass transfer coefficient constant, A crystal surface area, and c, c_i, and c^* are the solute concentrations in the bulk solution, at the interface and at equilibrium saturation.

Because c_i is a function of the mass transfer onto the crystal surface, the two equations can be linked together. If we assume the solvent does not experience any net mass transfer, c_i is governed by this differential equation:

$$\frac{dc_i}{dt} = \frac{dm_{\text{diffusion}}}{dt} - \frac{dm_{\text{deposition}}}{dt} = k_d A(c - c_i) - k_r A(c_i - c^*)^i$$

Since c_i can't easily be measured, it is common to simplify to this:

$$\frac{dm}{dt} = K_G(c - c^*)^s$$

In aqueous solutions, s is often between 1 and 2. K_G is related to k_d, k_r, s, and i, but it's probably better to just use K_G since again knowing c_i is difficult.

Interesting here that they do not use the saturation index or supersaturation ratio, but instead use the concentration directly.

5 Proposed Mass Model

I think we can use the textbook model to create a good A. The text book admits after it gives that model though that it is very simple and doesn't capture the size dependence of growth rate. By including that exponential term I had before I think we can capture that size dependence with a simple and intuitive time constant.

$$\frac{dM_{\text{scale}}}{dt} = K_G(c - c^*)^s(1 - e^{-t/\tau})$$

Often this equation ends up looking like one of the earlier equations depending on which stage is rate limiting. If diffusion is rate limiting, then $K_G = k_d A$ and $s = 1$. If deposition is rate limiting, then $K_G = k_r A$ and $s = i$.

But I do still feel the lack of a maximum scale mass is a major gap.

6 Problems

As I implement this model I see some problems. First, there is no behavior to keep scale mass from dropping below zero. However, I think an improved model where the $e^{-t/\tau}$ term is replaced with a size dependent term.

The second problem is we need a new mass balance equation for the solute. Initially I was building the scaling model into the membrane eqs block. However, after thinking some more, I think a better approach is to have a separate scaling eqs block. The mass transfer of the scaling block can be summed with the mass transfer of the membrane block, and the properties of the volume will be routed to both blocks. This not only makes the mass balance work out, but also allows for modeling scaling in pipes generally, so not just on RO membrane. Note that we will need to make area an input rather than a parameter in the membrane EQs block in this version.

The final problem is one I don't have a good solution for yet. We can easily* create a model for membrane area loss as a function of scale mass. However, when I had been thinking about this problem before, I was only thinking of situations where there is only scaling on one side of the membrane. Without knowing which side of the membrane the scaling is on, if we assume scaling on one side is minor, an easy way to deal with this is just to use max(area loss A, area loss B), or to keep things smooth, area loss area loss A + area loss B. However, if we have significant scaling on both sides of the membrane, this approach will over predict area loss. There is likely a stats solution here.

7 Scale size dependence model

I want to replace this part of the equation:

$$(1 - e^{-t/\tau})$$

with something that depends on scale size. A sigmoid function would be a good choice here. Something like this:

$$\frac{1}{1 + e^{-A(M_{\text{scale}} - B)}}$$

Trick is what are A and B. A is similar to our time constant from before. B is critical because we want this function to be near 0 when M_{scale} is near 0. But the number makes that work is dependent on A.

Alternatively, since setting smooth limits is actually already a function in simscape, we can just use scale area, A. Just throwing area into the equation makes the equation look more like those first equations from the textbook:

$$\frac{dM_{\text{scale}}}{dt} = k_g A(c - c^*)^s$$

where k_g switches between k_d and k_r depending on the limiting mass transfer mode.

For now, I make A a linear function of M_{scale} using a “specific area” term, a [m^2/kg].

$$A = aM_{\text{scale}}$$

However, there also needs to be some limits. First, the scale area can't exceed a maximum area (area of membrane or pipe surface). Second, if the mass of the scale is zero, this means there would never be growth. To prevent this, we set the minimum area to the nucleation area. These limits are set using a Simscape blend function for each limit.